



IMAGING EVIDENCE

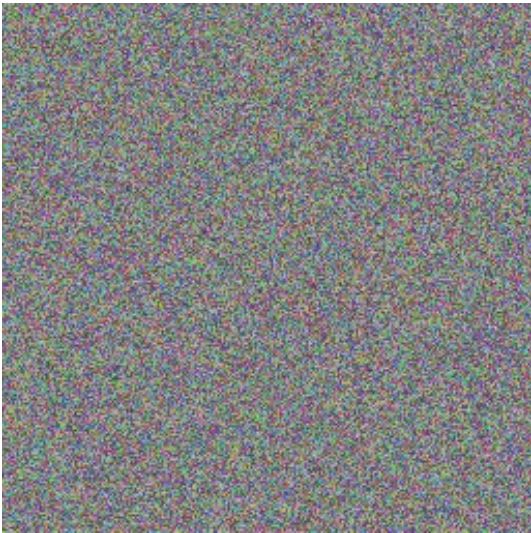


Figure 1: Original MRI scan

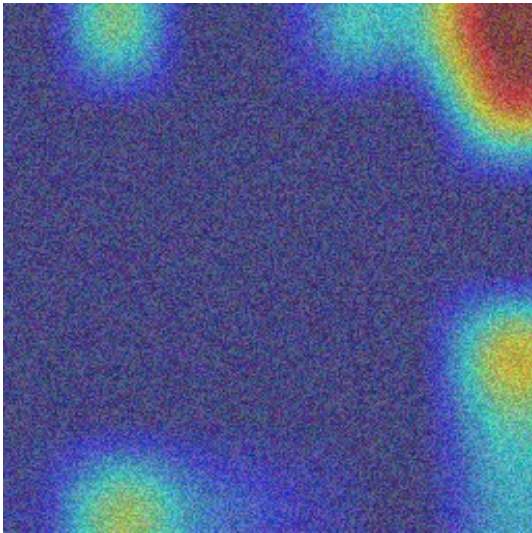


Figure 2: Grad-CAM activation overlay

GRAD-CAM ANALYSIS PANEL

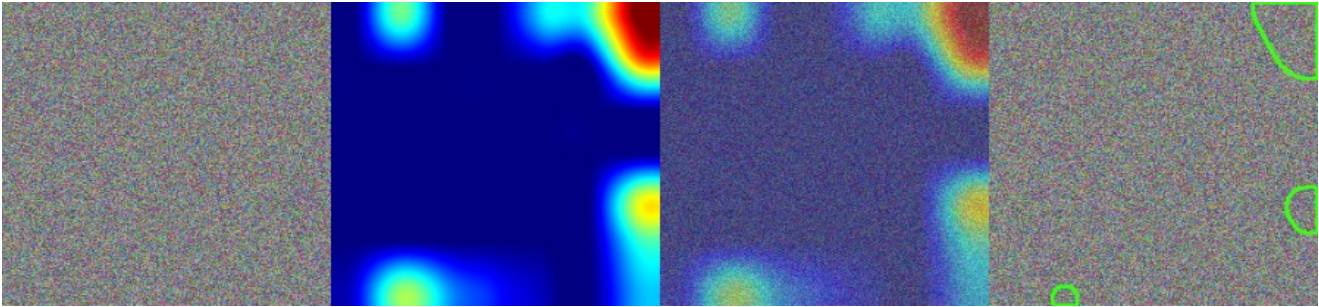


Figure 3: (Left→Right) Original MRI | Activation Map (Jet) | GradCAM Overlay | Tumour Boundary (threshold > 0.5)

■ Red areas = high tumour probability ■ Yellow = moderate ■ Blue = low activation

TUMOUR LOCALISATION (GRAD-CAM ANALYSIS)

| Parameter             | Value                | Notes                                       |
|-----------------------|----------------------|---------------------------------------------|
| Estimated Area        | 5.6% of visible scan | Low                                         |
| Hemisphere / Location | RIGHT hemisphere     | Based on centroid x-position                |
| Centroid Position     | (222, 78) pixels     | Hottest activation pixel centroid           |
| Bounding Region       | [49,0] → [255,255]   | Pixel coordinates of activation box         |
| Activation Score      | 73 / 100             | Mean activation in region (0=low, 100=high) |
| GradCAM Computed      | 0.52s                | CPU inference time                          |

## CLINICAL INTERPRETATION

The AI model identifies imaging features consistent with a glial neoplasm. The Grad-CAM localisation map highlights a region of interest with high activation intensity. Gliomas represent the most common primary brain tumours and vary widely in grade (WHO I–IV). Immediate referral to neurosurgery and neuro-oncology is strongly recommended. MRI with contrast enhancement, spectroscopy, and perfusion imaging should be obtained. Biopsy and molecular profiling (IDH status, MGMT methylation) may be required for definitive grading.

## RECOMMENDATIONS

- Urgent MRI brain with gadolinium contrast enhancement
- Neurosurgical consultation within 48 hours
- Neuro-oncology multidisciplinary team review
- Biopsy / resection planning (IDH, MGMT, EGFR profiling)
- Baseline neuropsychological assessment

## TECHNICAL DETAILS

| Parameter       | Value                                             |
|-----------------|---------------------------------------------------|
| Architecture    | EfficientNet-B3 + ResNet50+CBAM + DenseNet121     |
| Inference       | TTA (6-view: orig, H-flip, V-flip, rot90/180/270) |
| Ensemble method | Confidence-weighted average + temperature scaling |
| GradCAM method  | Standard Grad-CAM on last conv layer per backbone |
| Model AUC       | 0.9996 (macro one-vs-rest, test set n=1311)       |
| Model Accuracy  | 98.78%                                            |
| Sensitivity     | 98.68%   Specificity: 99.59%                      |
| Device          | CPU (no GPU detected)                             |
| Processing time | 1.20s (inference) + 0.52s (GradCAM)               |

■ **DISCLAIMER:** This report is generated by NeuroScan AI, an AI-assisted screening tool. It is intended for informational and research purposes only and does **NOT** constitute a medical diagnosis. All findings must be reviewed and confirmed by a qualified radiologist or neurologist before any clinical decision is made. Individual model performance may vary across patient populations and imaging protocols. Reported metrics: AUC 0.9996, Accuracy 98.78%, Sensitivity 98.68%, Specificity 99.59% (test set n=1,311 images). Report generated: 2026-04-11 12:26:01 IST | Report ID: RPT-95454859